



7081 - A Race Against the Clock: Too Much Nitrogen, Too Early?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	2	Plan-3shutter	NIRSpec MultiObject Spectroscopy	(2) Highz-CNO-Targets

ABSTRACT

Chemical abundances encode the integrated star formation over the lifetimes of galaxies. Since stars produce nitrogen and oxygen on different timescales, the relative abundance of N/O acts as a clock constraining the star formation history. Specifically, N/O describes the relative ratio of

moderately old stars (>100 Myr) to very young stars (<10 Myr). Recently, JWST observations of extreme star-forming galaxies only a few 100 Myrs after the big bang have found extreme UV N IV] and N III] emission lines. These lines suggestsuper-solar N/O ratios, implying a contradiction: galaxies existing for $<$ a few 100 Myr seem to have N enrichment from substantial older stellar populations. These galaxies have too much N too early in the universe. However, accurate relative abundances from rest-UV lines are extremely sensitive to the physical conditions assumed, such as the high densities ($> 1e6 \text{ cm}^{-3}$) measured for some high- z galaxies. Thus, unearthing the conditions that produce the N lines is essential to constrain the chemical enrichment of the first galaxies.

We propose 25.8 hours of high-resolution NIRSpec MSA observations of $z>5$ galaxies with existing detections of FUV NIV], NIII], OIII] and CIII] emission lines, providing the first sample of accurate measurements of high-ionization gas densities and N/O ratios derived under these conditions. This will allow us to test whether current models of rapid N enrichment are correct or if they are driven by extreme environments at high- z . The outcome will enable a blueprint to diagnose the properties of extreme N emission and reshape how we interpret metallicity and star formation in the earliest galaxies.

OBSERVING DESCRIPTION

We propose high-resolution NIRSpec MSA observations of high- z galaxies and/or AGN in GOODS-S. Our primary targets are 6 $z>5$ sources with significant rest-UV C, N, and O line detections in their archival medium and low-resolution spectra. We will simultaneously target a large number of $z>5$ galaxies whose potential CIII] 1907,1909 emission lines will be covered by the proposed spectra configuration. We will use NIRSpec MSA in its high resolution ($R\sim 2700$) mode with the G140H/F100LP for the goal of ensuring that the high-ionization density-sensitive CIII] 1907,1909 and NIV] 1483,1487 doublets are resolve. We calculate ETCs assuming the faintest individual CIII] line detection in our sample may only be a 20% of the integrated CIII] flux detected in the lower-resolution archival spectra.

Proposal 7081 - Targets - A Race Against the Clock: Too Much Nitrogen, Too Early?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	Highz-CNO-Targets	RA: 03 32 29.7199 (53.1238329d) Dec: -27 47 2.78 (-27.78411d) Equinox: J2000		
Comments: Description=[]					

Proposal 7081 - Observation 2 - A Race Against the Clock: Too Much Nitrogen, Too Early?

Tue Mar 11 20:04:47 GMT 2025

Observation	Proposal 7081, Observation 2: Plan-3shutter											Tue Mar 11 20:04:47 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(2)	Highz-CNO-Targets		RA: 03 32 29.7199 (53.1238329d) Dec: -27 47 2.78 (-27.78411d) Equinox: J2000								
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold	
	MSATA	false		After Target ACQ	MSA Center	Highz-CNO-Targets (13 sources)		jwst-nirspec-g140h		1.5		
Reference Stars												
Confirmation	#	Confirmation Type		Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time		
	1	After Target Acq		NRSIRS2	2	1		1		160.478		

Proposal 7081 - Observation 2 - A Race Against the Clock: Too Much Nitrogen, Too Early?

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140H/F100LP)	c1	3 Shutter Slitlet	53.145925458333 33 Degrees - 27.782526666666 683 Degrees	159.98973988695 4			3	12	24684.402
	2	1 (G140H/F100LP)	c1	3 Shutter Slitlet	53.145925458333 33 Degrees - 27.782526666666 683 Degrees	159.98973988695 4			3	12	24684.402
	3	1 (G140H/F100LP)	c1	3 Shutter Slitlet	53.145925458333 33 Degrees - 27.782526666666 683 Degrees	159.98973988695 4			3	12	24684.402
Special Requirements	MSA Planned Aperture PA 160.0000 to 160.0000 Degrees (V3 21.4254303 to 21.4254303)										